Exercises: Constrained Optimization Methods

Mathematical Foundations for Machine Learning

Projection onto Convex Sets

Exercise 1.1 [Easy]
Let $C = \{x \in \mathbb{R}^n \mid \|x\|_2 \leq 1\}$ be the unit Euclidean ball.

(a) Write down the optimization problem that defines the projection $P_C(z)$ of a point $z \in \mathbb{R}^n$ onto C.

(b) Derive the closed-form expression for $P_C(z)$ by considering the two cases $\|z\|_2 \leq 1$ and $\|z\|_2 > 1$ separately. smartly choose vector

(c) Verify that $P_C(z)$ is non-expansive, i.e. $\|P_C(x) - P_C(y)\|_2 \leq \|x - y\|_2$, for the special case $\|x\|_2 > 1$ and $\|y\|_2 \leq 1$.

Exercise 1.2 [Medium]
Let $C = \{x \in \mathbb{R}^n \mid Ax = b\}$ where $A \in \mathbb{R}^{p \times n}$ with $\operatorname{rank}(A) = p < n$.

(a) Using the method of Lagrange multipliers, derive that the projection of a point $z \in \mathbb{R}^n$ onto C is given by

$$P_C(z) = z + A^T(AA^T)^{-1}(b - Az).$$

(b) Show that the projection matrix $P = I - A^T(AA^T)^{-1}A$ is idempotent, i.e. $P^2 = P$, and symmetric.

(c) Verify that for any $z \in \mathbb{R}^n$, the residual $z - P_C(z)$ is orthogonal to the affine subspace C in the sense that $\langle z - P_C(z), v\rangle = 0$ for all $v \in \ker(A)$.

Exercise 1.3 [Medium]
Let $C = \{x \in \mathbb{R}^n \mid x_i \geq 0,\ \sum_{i=1}^n x_i = 1\}$ be the probability simplex.

(a) Show that C is a convex set.

(b) A point $z = (0.5, -0.3, 0.2, 0.9)^T \in \mathbb{R}^4$ needs to be projected onto the simplex $C \subset \mathbb{R}^4$. Formulate the KKT conditions for the projection problem $\min_{x \in C} \|z - x\|_2^2$.

(c) Using the KKT conditions, argue that the solution has the form $x_i = \max\{z_i - \mu, 0\}$ for some $\mu \in \mathbb{R}$, and find the projection $P_C(z)$.

Exercise 1.4 [Hard]
Let $C \subseteq \mathbb{R}^n$ be a closed convex set and P_C the projection operator onto C.

(a) Prove the variational characterization: $x^* = P_C(z)$ if and only if

$$\langle y - x^*, z - x^*\rangle \leq 0 \quad \forall\, y \in C.$$

(Hint: Consider the function $g(t) = \|z - (x^* + t(y - x^*))\|_2^2$ for $t \in [0, 1]$.)

(b) Using part (a), prove that the projection is firmly non-expansive:

$$\|P_C(x) - P_C(y)\|_2^2 \leq \langle P_C(x) - P_C(y),\, x - y\rangle.$$

(c) Deduce from (b) that P_C is non-expansive: $\|P_C(x) - P_C(y)\|_2 \leq \|x - y\|_2$.

Projected Gradient Descent

Exercise 2.1 [Easy]
Consider the constrained optimization problem

$$\min_{x \in C} f(x) = \frac{1}{2} x^T Q x - c^T x,$$

where $Q = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$, $c = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, and $C = \{x \in \mathbb{R}^2 \mid x_1 \geq 0,\ x_2 \geq 0\}$ is the non-negative orthant.

(a) Compute the gradient $\nabla f(x)$ and the Lipschitz constant L of the gradient.

(b) Starting from $x^{(0)} = (0,0)^T$ with step size $t = 1/L$, perform two iterations of projected gradient descent:

$$x^{(k+1)} = P_C\big(x^{(k)} - t\,\nabla f(x^{(k)})\big).$$

(c) What is the unconstrained minimizer? Is it equal to the constrained minimizer? Why or why not?

Exercise 2.2 [Medium]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex with L-Lipschitz continuous gradient, and let C be a closed convex set. Consider projected gradient descent with constant step size $t = 1/L$.

(a) State the optimality condition for $x^* = \arg\min_{x \in C} f(x)$ in terms of the projection P_C and the gradient ∇f.

(b) Show that if $x^* = P_C(x^* - t\,\nabla f(x^*))$, then $\langle \nabla f(x^*), y - x^* \rangle \geq 0$ for all $y \in C$. Interpret this geometrically.

(c) Let $f(x) = \|x - p\|_2^2$ for some $p \notin C$. Show that projected gradient descent with step size $t = 1/L$ converges to the projection $P_C(p)$ in a single step from any $x^{(0)} \in C$. What is L in this case?

The angle between (z-x*) and (y-x*) is >= 90 -> cos() <= 0 -> when x* = p(z) = z (inside) then non defined angle -> inner product 0

Exercise 2.3 [Hard]
Consider the constrained LASSO problem:

$$\min_{x \in \mathbb{R}^n} \frac{1}{2}\|Ax - b\|_2^2 \quad \text{s.t.} \quad \|x\|_1 \leq \tau,$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $\tau > 0$.

(a) Show that the objective $f(x) = \frac{1}{2}\|Ax - b\|_2^2$ has Lipschitz continuous gradient with constant $L = \|A^T A\|_2 = \sigma_{\max}(A)^2$.

(b) Write down the projected gradient descent update for this problem. Which subproblem needs to be solved at each step?

(c) The projection onto the ℓ_1-ball $C = \{x \mid \|x\|_1 \leq \tau\}$ can be computed via soft-thresholding: $(P_C(z))_i = \operatorname{sign}(z_i)\max\{|z_i| - \lambda, 0\}$ where $\lambda \geq 0$ is chosen so that $\|P_C(z)\|_1 = \tau$ (or $\lambda = 0$ if $\|z\|_1 \leq \tau$). Prove that this operator is well-defined, i.e. that there exists a unique $\lambda \geq 0$ such that $\sum_i \max\{|z_i| - \lambda, 0\} = \tau$ whenever $\|z\|_1 > \tau$.

(d) State the convergence rate of projected gradient descent for this problem and compare it to the accelerated variant.

Interior Point / Barrier Method

$$\phi(x) = -\sum_{i=1}^{m} \log(-g_i(x))$$

Exercise 3.1 [Easy]
Consider the linear program

to get rid of the inequalities -> log cannot be 0, as you approach it penalty goes to 0 -> -sum then to infinity. -g(x) so log inside is not negative

$$\min_{x \in \mathbb{R}^2} c^T x \quad \text{s.t.} \quad x_1 \geq 0,\ x_2 \geq 0,\ x_1 + x_2 \leq 1,$$

so you take t * f(x) + (x) and so t gives importance to f, but unless you put t = infi

with $c = (1, 2)^T$.

(a) Write down the logarithmic barrier function $\phi(x)$ for the inequality constraints.

(b) Formulate the barrier problem: $\min_x \; t\, c^T x + \phi(x)$ (there are no equality constraints here).

(c) For a given $t > 0$, find the central point $x^*(t)$ by solving the first-order optimality conditions of the barrier problem.

(d) Verify that $x^*(t) \to (0, 0)^T$ (the LP optimum) as $t \to \infty$. What is the duality gap $f(x^*(t)) - p^*$ in terms of t?

p* true optimal value -> f(x*(t)) - p* is the extra cost bec. t not infinity

$$f(x^*(t)) - p^* \leq \frac{m}{t}$$

Exercise 3.2 [Medium]

Consider the general barrier method for the problem

$$\min_x f(x) \quad \text{s.t.} \quad g_i(x) \leq 0,\; i = 1, \ldots, m, \quad Ax = b.$$

(a) Define the logarithmic barrier function $\phi(x)$ and write the barrier subproblem for parameter $t > 0$.

(b) Let $x^*(t)$ be the solution of the barrier subproblem. Show that the KKT conditions of the barrier subproblem yield dual-feasible points for the original problem. Specifically, define $\lambda_i^*(t) = -\frac{1}{t\, g_i(x^*(t))}$ and show $\lambda_i^*(t) \geq 0$.

interior point method -> differentiate -> stationarity condition with dual vars

(c) Show that $f(x^*(t)) - p^* \leq \frac{m}{t}$, where p^* is the optimal value of the original problem.

(d) Interpret the complementary slackness violation: show $\lambda_i^*(t) \cdot (-g_i(x^*(t))) = \frac{1}{t}$ and explain what happens as $t \to \infty$.

Exercise 3.3 [Hard]

Consider the inequality-constrained quadratic program:

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^T Q x + c^T x \quad \text{s.t.} \quad a_i^T x \leq b_i,\; i = 1, \ldots, m,$$

where $Q \succ 0$.

(a) Write the barrier subproblem for parameter $t > 0$ and compute its gradient and Hessian with respect to x.

(b) Suppose $n = 1$, $Q = (2)$, $c = (-3)$, with a single constraint $x \leq 2$. Solve for the central path $x^*(t)$ explicitly as a function of t, and verify that $x^*(t) \to 3/2$ as $t \to \infty$.

(c) In the barrier method, one increases t by a factor $\gamma > 1$ at each outer iteration. Explain the trade-off in choosing γ: what happens for very small γ vs. very large γ?

(d) For the general case, explain how Newton's method would be applied to solve each barrier subproblem. Write down the Newton step Δx in terms of the Hessian and gradient of the barrier objective.

Quadratic Programs with Linear Equality Constraints

Exercise 4.1 [Easy]

Consider the equality-constrained quadratic program:

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^T Q x + c^T x \quad \text{s.t.} \quad Ax = b,$$

where $Q \succ 0$, $A \in \mathbb{R}^{p \times n}$ with $\operatorname{rank}(A) = p$.

(a) Write down the Lagrangian and derive the KKT system (a linear system in x and the multiplier ν).

(b) Show that the KKT system can be written as

$$\begin{pmatrix} Q & A^T \\ A & 0 \end{pmatrix} \begin{pmatrix} x \\ \nu \end{pmatrix} = \begin{pmatrix} -c \\ b \end{pmatrix}.$$

(c) Solve the system for $Q = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$, $c = \begin{pmatrix} -4 \\ -6 \end{pmatrix}$, $A = \begin{pmatrix} 1 & 1 \end{pmatrix}$, $b = 3$.

Exercise 4.2 [Medium]
Consider the problem of finding the minimum-norm solution:

$$\min_{x \in \mathbb{R}^n} \frac{1}{2}\|x\|_2^2 \quad \text{s.t.} \quad Ax = b,$$

where $A \in \mathbb{R}^{p \times n}$ with $\operatorname{rank}(A) = p$ and $p < n$.

(a) Derive the closed-form solution $x^* = A^T(AA^T)^{-1}b$ from the KKT conditions.

(b) Show that this is the Moore–Penrose pseudoinverse solution: $x^* = A^\dagger b$ where $A^\dagger = A^T(AA^T)^{-1}$ for full row-rank A.

(c) In the context of linear regression with more features than data points ($n > N$), this corresponds to the minimum-norm interpolation. If $A = X \in \mathbb{R}^{N \times n}$ and $b = y \in \mathbb{R}^N$, express the solution $\hat{\theta}$ and explain why this connects to the unregularized least-squares solution in the overparameterized regime.

(d) Show that x^* lies in the row space of A, i.e. $x^* \in \operatorname{range}(A^T)$.

Exercise 4.3 [Hard]
Consider the equality-constrained quadratic program with a general positive semidefinite matrix:

$$\min_{x \in \mathbb{R}^n} \frac{1}{2}x^T Q x + c^T x \quad \text{s.t.} \quad Ax = b,$$

where $Q \succeq 0$ (PSD but possibly singular), $A \in \mathbb{R}^{p \times n}$, $\operatorname{rank}(A) = p$.

(a) The KKT matrix $K = \begin{pmatrix} Q & A^T \\ A & 0 \end{pmatrix}$ may be singular when Q is only PSD. Give a necessary and sufficient condition on Q and A for K to be nonsingular (Hint: consider $\ker(Q) \cap \ker(A)$).

(b) When K is nonsingular, derive the Schur complement formula: eliminating x from the KKT system yields the reduced system $AQ^{-1}A^T\nu = AQ^{-1}c + b$ (assuming $Q \succ 0$). When is this approach computationally preferable?

(c) Apply part (b) to the Ridge regression problem: $\min_\theta \frac{1}{2}\|X\theta - y\|^2 + \frac{\alpha}{2}\|\theta\|^2$ subject to $e^T\theta = 0$ (zero-mean constraint), where $e = (1, \ldots, 1)^T$. Derive the constrained Ridge estimator.

Penalty Methods and Accelerated Gradient Descent

Exercise 5.1 [Easy]
Consider the equality-constrained problem:

$$\min_{x \in \mathbb{R}^2} x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 = 1.$$

(a) The quadratic penalty method replaces this with the unconstrained problem

$$\min_x f(x) + \frac{\rho}{2}\|h(x)\|^2 = x_1^2 + x_2^2 + \frac{\rho}{2}(x_1 + x_2 - 1)^2.$$

Find the minimizer $x^*(\rho)$ for a given $\rho > 0$.

(b) Show that $x^*(\rho) \to (1/2, 1/2)^T$ as $\rho \to \infty$, and verify that $(1/2, 1/2)^T$ is the constrained minimizer.

(c) Explain why, in practice, one cannot simply use a very large ρ from the start.

Exercise 5.2 [Medium]
Consider Nesterov's accelerated projected gradient descent for $\min_{x \in C} f(x)$ where f has L-Lipschitz continuous gradient and C is closed convex. The iterations are:

$$x^{(k+1)} = P_C\left(y^{(k)} - \frac{1}{L}\nabla f(y^{(k)})\right), \qquad y^{(k+1)} = x^{(k+1)} + \frac{k}{k+3}\left(x^{(k+1)} - x^{(k)}\right),$$

with $y^{(0)} = x^{(0)} \in C$.

(a) Compare the convergence rates of projected gradient descent and accelerated projected gradient descent. Express both in terms of the number of iterations needed to achieve $f(x^{(k)}) - f^* \leq \varepsilon$.

(b) The momentum coefficient is $\beta_k = \frac{k}{k+3}$. What is $\lim_{k\to\infty} \beta_k$? Why does this not cause divergence?

(c) Consider $f(x) = \frac{1}{2}x^T Q x$ with $Q = \mathrm{diag}(1, 100)$ and $C = \mathbb{R}^2$ (unconstrained). Starting from $x^{(0)} = (100, 100)^T$, explain qualitatively why accelerated gradient descent converges faster than standard gradient descent for this problem.

Exercise 5.3 [Hard]
Consider the inequality-constrained problem:

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad g_i(x) \leq 0,\ i = 1, \ldots, m.$$

(a) The **exterior penalty method** uses the penalized objective $P_\rho(x) = f(x) + \frac{\rho}{2}\sum_{i=1}^m \max\{0, g_i(x)\}^2$. Explain why, for finite ρ, the minimizer $x^*(\rho)$ typically violates the constraints. Contrast this with the barrier method.

(b) Prove that if f and all g_i are convex, then P_ρ is also convex.

(c) Consider $\min_{x \in \mathbb{R}} x^2$ subject to $x \geq 1$ (equivalently $-x + 1 \leq 0$). Solve the penalty subproblem $\min_x x^2 + \frac{\rho}{2}\max\{0, 1 - x\}^2$ analytically, find $x^*(\rho)$, and show $x^*(\rho) \to 1$ as $\rho \to \infty$.

(d) Compare the three approaches for handling constraints: penalty method, barrier method, and projected gradient descent. Discuss when each is most appropriate.

Mixed Challenging Exercises

Exercise 6.1 [Hard]
Consider training a linear classifier with a margin constraint. Given data points (a_i, y_i) with $a_i \in \mathbb{R}^n$ and $y_i \in \{-1, +1\}$ for $i = 1, \ldots, m$, we solve:

$$\min_{w \in \mathbb{R}^n} \frac{1}{2}\|w\|_2^2 \quad \text{s.t.} \quad y_i(a_i^T w) \geq 1,\ i = 1, \ldots, m.$$

(a) This is a quadratic program with linear inequality constraints. Write the Lagrangian and derive the KKT conditions.

(b) Formulate the barrier subproblem for parameter $t > 0$. Write the gradient of the barrier objective.

(c) Alternatively, one could solve this via projected gradient descent. The constraint set is $C = \{w : y_i a_i^T w \geq 1, \forall i\}$. Is C convex? Is the projection onto C easy to compute in general?

(d) Based on your answers, argue which method (barrier vs. projected GD vs. penalty) is most appropriate for this problem.

Exercise 6.2 [Hard]
Consider the entropy-regularized optimal transport problem in 1D. Given two discrete distributions $p, q \in \mathbb{R}^n$ on the probability simplex, and a cost matrix $C \in \mathbb{R}^{n\times n}$, we solve:

$$\min_{X\in\mathbb{R}^{n\times n}} \langle C, X\rangle + \frac{1}{t}\sum_{i,j} X_{ij}\log X_{ij}$$

$$\text{s.t.} \quad X\mathbf{1} = p, \quad X^T\mathbf{1} = q, \quad X_{ij} \geq 0.$$

(a) Show that the objective is strictly convex in X for $t > 0$ and $X_{ij} > 0$.

(b) Write the Lagrangian (ignoring the non-negativity constraints, as the entropy term ensures $X_{ij} > 0$ at the optimum). From the KKT stationarity condition, show that the optimal X^* has the form $X^*_{ij} = u_i K_{ij} v_j$ for some vectors $u, v > 0$ and matrix $K_{ij} = e^{-tC_{ij}}$.

(c) The problem in (b) can be seen as the equality-constrained optimization on the positive orthant. Explain why the barrier method is unnecessary here — the entropy regularization already acts as a barrier.

(d) If instead we dropped the entropy term and solved the pure LP with the simplex constraints, explain how the interior point method would approach this problem.

Exercise 6.3 [Hard]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be μ-strongly convex with L-Lipschitz continuous gradient, and let C be a closed convex set.

(a) Show that the constrained problem $\min_{x\in C} f(x)$ has a unique minimizer x^*.

(b) For projected gradient descent with step size $t = 1/L$, prove the descent lemma:

$$\|x^{(k+1)} - x^*\|^2 \leq \left(1 - \frac{\mu}{L}\right)\|x^{(k)} - x^*\|^2.$$

(Hint: Use strong convexity, the Lipschitz gradient condition, and the non-expansiveness of P_C.)

(c) How many iterations does projected gradient descent need to achieve $\|x^{(k)} - x^*\|^2 \leq \varepsilon$ as a function of $\kappa = L/\mu$ (the condition number)? Compare with the accelerated rate.

(d) Relate the condition number κ to a concrete ML setting: for Ridge regression $\min_{\|\theta\|_1\leq\tau} \frac{1}{2}\|X\theta - y\|^2 + \frac{\alpha}{2}\|\theta\|^2$, what is κ?